

Assignment Title	Smart Traffic Junction Controller
Learning outcomes covered	LO1 to LO5
Assignment Mode	Group assignment (2 students per group)
Maximum Marks	100
Contribution to the Final Grade	30%
Date published	
Deadline for submissions	4 weeks from the date of assignment release
Mode of Submission	Final report (PDF), source code (.zip) and schematics (PDF) submitted via LMS

Assignment Objective

The goal of this project is to design and build a scaled model of a four-way traffic junction controlled entirely by an Arduino Uno programmed in AVR Assembly language. The system must manage traffic flow across four approaches (North, South, East, West) using timed signal cycling in Normal mode, and switch to a flashing yellow Night mode based on ambient light sensing. A pedestrian crossing on the South approach must respond to a push-button request by safely halting all vehicle traffic and providing a 5-second countdown for pedestrians to cross. Upon completion, students will have demonstrated the ability to implement multi-port I/O control, timer-driven sequencing, interrupt-based event handling, analog sensor reading, and real-time countdown display entirely in AVR Assembly language.

Scenario

Design and implement a scaled model of a four-way traffic junction controlled entirely by an Arduino Uno programmed in AVR Assembly. The junction has four approaches (North, South, East, West), each with a set of three LEDs (red, yellow, green) representing traffic lights. The system must operate in two modes: NORMAL (timed cycle through all four directions) and NIGHT MODE (triggered by an LDR connected to the ADC, when ambient light drops below a threshold, all directions flash yellow at 1 Hz).

A pedestrian crossing is located on the South approach of the junction, between the junction center and the South-side pavement. A push-button is mounted at this crossing point. When a pedestrian presses the button, the system waits for the current vehicle phase to complete, then forces all four directions to red and grants a 5-second pedestrian crossing window. During this window, a 7-segment display (or a row of 5 LEDs that turn off one-by-one) counts down from 5 to 0, giving the pedestrian a visible indication of remaining crossing time. Once the countdown expires, normal traffic cycling resumes from where it left off.

The pedestrian request must be deferred, pressing the button during any active green or yellow phase only sets a request flag; the crossing phase is inserted only after the current vehicle phase and its all-red clearance interval have completed.

Components

Students are expected to identify and source all relevant components required to build and operate the traffic junction model based on the scenario described above. No components are provided by the lab. At minimum, the system will require an Arduino Uno (ATmega328P) as the microcontroller, 12 LEDs with current-limiting resistors for the four sets of traffic lights (red, yellow, green per direction), a momentary push-button for the pedestrian crossing on the South approach, either a 7-segment display or a row of 5 LEDs for the pedestrian countdown, an LDR with a voltage divider for ambient light sensing, a battery pack for power, and a base board (approximately 60 cm x 60 cm) to mount the junction model. Students may include additional components such as a pedestrian crossing indicator LED or mode indicator LEDs where justified.

All component selections, resistor values, pin-to-port mappings, and wiring decisions must be documented and justified in the final report with a complete schematic diagram.

⚠ Important: All firmware must be written exclusively in AVR Assembly language. The use of C, C++, or Arduino library functions is not permitted.

Physical Demonstration

All groups are required to present a live demonstration of their working traffic junction model during the scheduled demonstration session. The demonstrator and module lecturer will assess the system's performance across all operating modes in sequence. The junction must be built on a board or thick cardboard (approximately 60 cm x 60 cm) with four road approaches clearly marked converging at the center. LED sets must be visibly mounted at each approach, the pedestrian push-button and countdown display must be positioned on the South approach, and the LDR must be mounted in an accessible location where the assessor can easily cover it to simulate nightfall. The following describes what is expected during each stage of the demonstration.

Mode 1: Normal Traffic Cycle

The system will be powered on and placed in Normal mode. The assessor will observe the traffic lights cycling through the full four-way sequence: North-South green with East-West red, transitional yellow, all-red clearance, then East-West green with North-South red, and so on. The assessor will verify that phase durations match the timing values defined in the group's

program memory table, that transitions follow the correct order without skipping or repeating phases, and that the all-red clearance interval appears between every directional change.

Mode 2: Pedestrian Crossing

While the junction is running in Normal mode, the assessor will press the pedestrian push-button on the South approach at an arbitrary point during a green or yellow phase. The assessor will verify that the traffic lights do not change immediately upon the button press: the current vehicle phase must complete its remaining duration and the all-red clearance must occur before the pedestrian crossing phase is inserted. Once the crossing phase begins, all four directions must show red, and the countdown display (7-segment or LED bar) must count down visibly from 5 to 0 at one-second intervals. The assessor will confirm that after the countdown expires, normal traffic cycling resumes from the correct next phase. The assessor will also press the button multiple times in quick succession to verify that only a single crossing phase is granted per request, and will press the button during the crossing phase itself to confirm that no additional crossing is queued until the current one completes and at least one vehicle phase has passed.

Mode 3: Night Mode

The assessor will cover the LDR with a hand or opaque material to simulate a drop in ambient light. The assessor will observe whether the system transitions from Normal mode into Night mode, where all four yellow LEDs flash simultaneously at a 1 Hz rate and all other LEDs are off. The assessor will then uncover the LDR and verify that the system does not immediately snap back to Normal mode: the hysteresis mechanism must require the light level to remain above the threshold for five consecutive seconds before Normal mode resumes. The assessor will briefly uncover and re-cover the LDR within the five-second window to confirm that the hysteresis counter resets correctly and Night mode is maintained. If the pedestrian button is pressed during Night mode, the system should either ignore the request or defer it until Normal mode resumes: the chosen behavior must be documented in the report.

Deliverables

Final Report

- Detailed report covering hardware design, firmware architecture, implementation, testing results, and discussion.
- Must include: complete schematic diagram, state machine diagram showing all modes and transitions, subroutine flowcharts for key routines (phase cycling, pedestrian handling, night mode detection, countdown display), pin-to-port mapping table, phase duration tables, test result tables, and photographs of the completed junction model from multiple angles.
- File naming: IE3064_Assignment_Group_X.pdf

Presentation & Viva

- Slide deck summarizing hardware design, assembly firmware structure, mode operation, and test results.
- Each group member must be able to explain any part of the system independently during the viva, including the assembly-level implementation of each module (timer configuration, interrupt handling, ADC reading, countdown logic, and port manipulation).

Source Code Submission

- All .asm source files submitted as a .zip via LMS.
- Code must assemble without errors using AVR-GCC (avr-as) on a clean installation.
- Include a README.md with assembly toolchain setup instructions and a description of each source file.

Assignment Guidelines

Group Composition

- 2 students per group.
- Each member must be clearly assigned to specific modules in the report (e.g., one member responsible for traffic light cycling and night mode, the other for pedestrian crossing and countdown display). Individual contribution statements must be included in the report.

Duration

- 4 weeks from the date of assignment release.

File Naming Convention

- Final report: IE3064_Assignment_Group_X.pdf
- Source code: IE3064_Group_X_Code.zip
- Schematics: IE3064_Group_X_Schematics.pdf

Firmware Constraint

- All firmware must be written entirely in AVR Assembly language. Using any language other than AVR Assembly will nullify all marks for the affected component.

Academic Integrity

- All submitted work must be original. Code or designs copied from external sources without proper attribution will be treated as plagiarism under university academic integrity policy. Students may reference datasheets, AVR instruction set manuals, and lecture materials, but all implementation must be their own.

Marking Scheme

⚠ Using any language other than AVR Assembly language will nullify all marks for the affected component.

Criteria	Marks (%)
Testing & Performance Evaluation	40
Normal Mode — correct four-way phase sequence with accurate timing	10
Normal Mode — proper transitional yellow and all-red clearance intervals	5
Pedestrian Crossing — deferred handling (no immediate light change on button press)	5
Pedestrian Crossing — all directions held red during crossing phase	3
Pedestrian Crossing — countdown display (7-segment or LED bar) counts 5 to 0 correctly	5
Pedestrian Crossing — multiple presses grant only one crossing phase	2
Pedestrian Crossing — normal cycling resumes from correct phase after crossing	2
Night Mode — yellow LEDs flash at 1 Hz when LDR is covered	4
Night Mode — hysteresis mechanism prevents immediate exit (5 consecutive seconds above threshold)	4
Report Quality & Documentation	25
Complete and accurate schematic diagram with pin-to-port mapping	5
State machine diagram covering all modes and transitions	4
Subroutine flowcharts for key firmware routines	4
Component selection and justification	3

Phase duration tables and test result tables	3
Overall report structure, clarity, and presentation (including photographs)	3
Individual contribution statements	3
Presentation Quality and Skills	15
Clear and well-structured slide deck	5
Effective explanation of hardware design and firmware architecture	5
Demonstration of test results and mode operation during presentation	5
Viva / Individual Understanding	20
Ability to explain register-level timer, interrupt, and ADC configuration	5
Ability to explain traffic phase sequencing and state machine logic	5
Ability to explain pedestrian crossing handling and countdown implementation	5
Ability to explain night mode detection, hysteresis, and mode transitions	5
Total	100